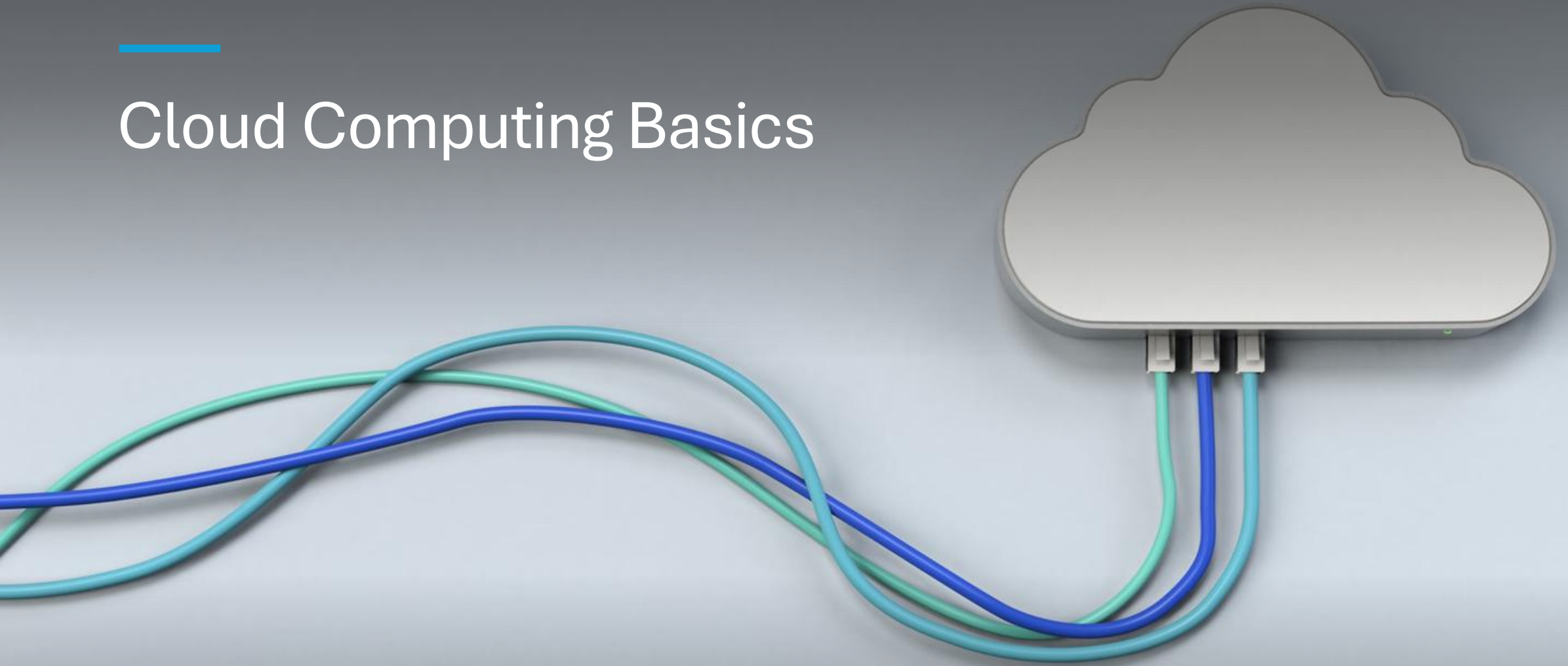

Cloud Computing Basics



What is Cloud Computing?

- Method for accessing technology services (e.g., storage, databases, computing power) via the Internet
- Eliminates the need for owning or maintaining physical hardware
- Allows users to store data and run applications remotely
- Operates on a pay-as-you-go pricing model — pay only for what you use
- Provides flexibility to scale resources up or down as needed
- Delivers cost-effective and efficient solutions for individuals and businesses

Cloud Deployment Models

- Cloud computing can be used by individuals, companies, or teams based on needs
- Not all clouds operate the same — each is designed for different scenarios
- Three main cloud deployment models:
 - Public Cloud
 - Private Cloud
 - Hybrid Cloud



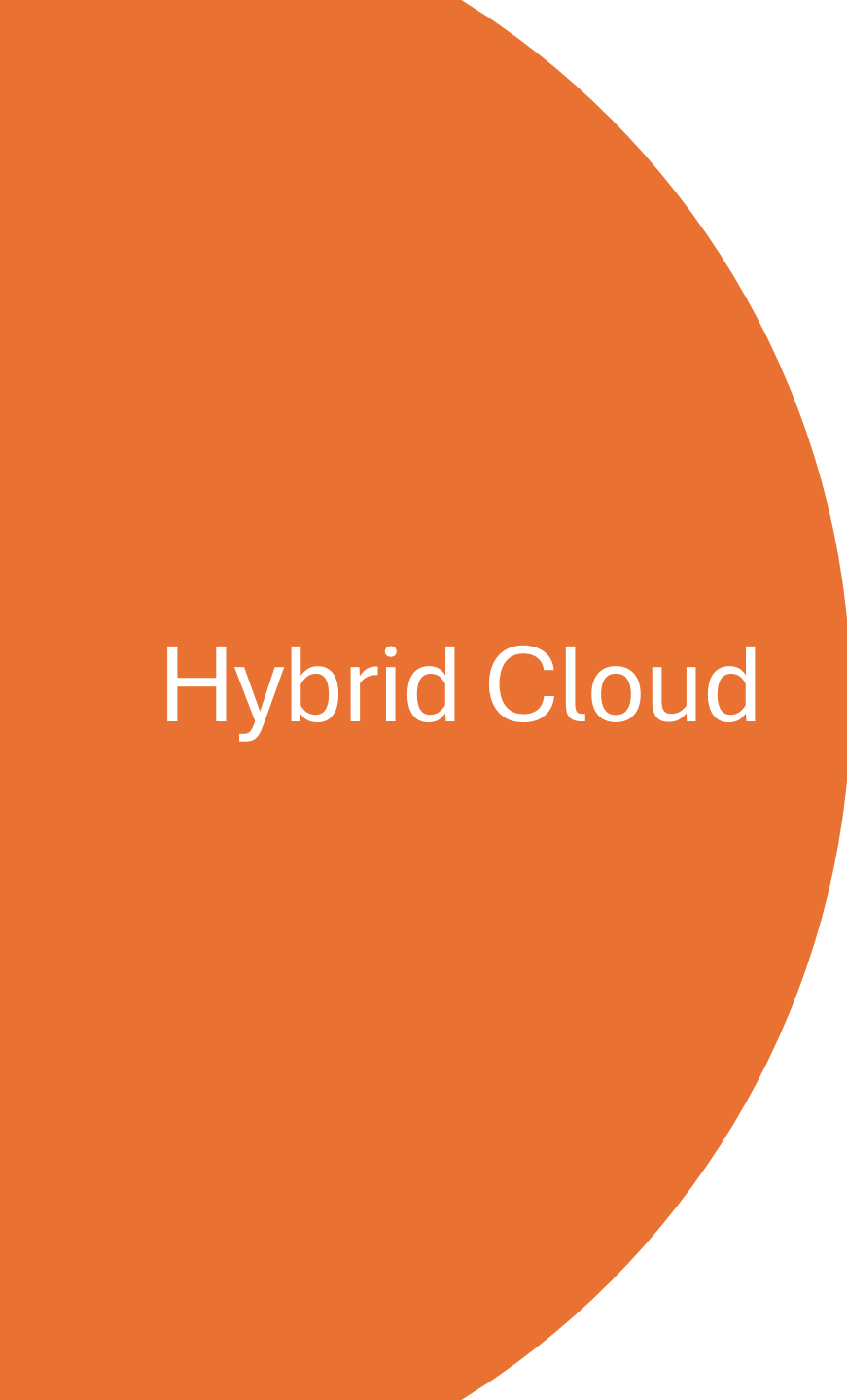
Public Cloud

- Services delivered over the Internet
- Resources shared among multiple organizations
- Accessible to anyone who wants to use the services
- Cost-effective and widely available

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Private Cloud

- Used exclusively by a single organization
 - Hosted on a private network
 - Provides greater control, security, and data privacy
 - Infrastructure not shared with others
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Hybrid Cloud

- Combination of public and private cloud environments
 - Sensitive data stored in private cloud
 - Less-critical operations run on public cloud
 - Offers flexibility, optimized performance, and cost efficiency
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What is AWS?

- AWS (Amazon Web Services) is a cloud computing platform provided by Amazon.
- Offers a wide range of on-demand technology services delivered over the Internet.
- Trusted and widely adopted by organizations of all sizes globally.
- Operates using a global network of secure, scalable data centers.

AWS Key Features

- Over 200+ fully featured services including:
 - Computing power (e.g., EC2)
 - Storage (e.g., S3)
 - Databases (e.g., RDS, DynamoDB)
 - Networking
 - Security tools
 - Machine learning and AI services
 - Developer tools
 - Supports almost any type of application or workload.

AWS Key Features...

- Global Infrastructure
 - AWS has a global footprint with data centers across multiple continents.
 - Organized into:
 - Regions (geographical areas)
 - Availability Zones (AZs) (multiple isolated data centers within a region)
 - Ensures high availability, fault tolerance, and low latency.

AWS Key Features...

- Pay-as-You-Go Pricing
 - Users only pay for what they use—no upfront hardware costs.
 - Helps businesses reduce IT expenditure and scale efficiently.



AWS Key Features...

- Highly Scalable and Flexible
 - Easily scale resources up or down based on demand.
 - Ideal for startups, large enterprises, and anyone needing flexible computing power.



AWS Key Features...

- Security and Compliance
 - Provides strong security features:
 - Encryption
 - Identity and access management
 - Network firewalls
 - Meets global compliance standards—trusted in finance, healthcare, and government sectors.



AWS Key Features...

- Widely Adopted. Used by millions of customers including:
 - Startups
 - Enterprise companies
 - Government agencies
 - Educational institutions



Core AWS Services Overview

- AWS offers essential services across compute, storage, networking, and databases.
- Understanding these core services helps you build scalable, efficient cloud solutions.
- These are foundational for AWS beginners and often covered in AWS certification training.

AWS Compute Services

- Amazon EC2 (Elastic Compute Cloud)
 - Provides resizable virtual servers for flexible compute capacity.
 - Supports scaling resources up or down based on workload needs.
 - Ideal for developers building web-scale applications.
- AWS Lambda
 - Serverless compute service — run code without managing servers.
 - Automatically handles scaling and infrastructure management.
 - Simplifies application deployment and reduces operational overhead.
- Importance:
 - Essential for beginners learning how to run and manage cloud applications efficiently.



AWS Storage & Networking Services

- AWS Storage
 - Amazon S3
 - Highly reliable, scalable storage for files and large datasets.
 - Secure, durable, and accessible from anywhere.
- Amazon EBS
 - Persistent block storage for EC2 instances.
 - Ensures data durability and high availability for running applications.
- AWS Networking
 - Amazon VPC
 - Creates a private, isolated network within AWS.
 - Provides control over subnets, route tables, access rules, and gateways.
- Amazon Route 53
 - Highly available DNS service.
 - Routes internet traffic to your applications reliably

AWS Database Services

- Amazon RDS (Relational Database Service)
 - Simplifies setup, operation, and scaling of relational databases.
 - Designed for structured data with minimal management required.
- Amazon DynamoDB
 - Fully managed NoSQL database service.
 - Fast, flexible, and supports automatic scaling for high-traffic apps.
- Importance:
 - Critical for Cloud Practitioners to manage and support application data efficiently.



Understanding AWS Pricing & Billing

- AWS offers flexible pricing options to suit different workloads and budgets.
 - Supports cost optimization through free tiers, pricing models, and cost management tools.
 - Essential knowledge for Cloud Practitioner and AWS beginners.
- 

AWS Free Tier

- Provides access to many AWS services at no cost for the first 12 months.
- Includes usage allowances for compute, storage, databases, and more.
- Ideal for learning, testing, and experimenting with AWS services.
- Helps beginners explore AWS without upfront expenses.

AWS Pricing Models

- On-Demand
 - Pay per hour/second based on actual usage.
 - No long-term commitments.
 - Ideal for unpredictable or short-term workloads.
- Reserved Instances
 - Commit to 1-year or 3-year usage for significant discounts (up to 75%).
 - Best for stable, long-term workloads.
- Spot Instances
 - Use unused EC2 capacity at reduced prices.
 - Highly cost-effective but workloads may be interrupted.
 - Suitable for flexible, fault-tolerant jobs (batch processing, data analysis).

AWS Cost Management Tools

- AWS Budgets
 - Set custom spending limits.
 - Receive alerts when approaching or exceeding thresholds.
 - Helps maintain financial control.
- AWS Cost Explorer
 - Visualize cost and usage trends over time.
 - Identify spending patterns and optimization opportunities.
 - Useful for forecasting and cost planning.

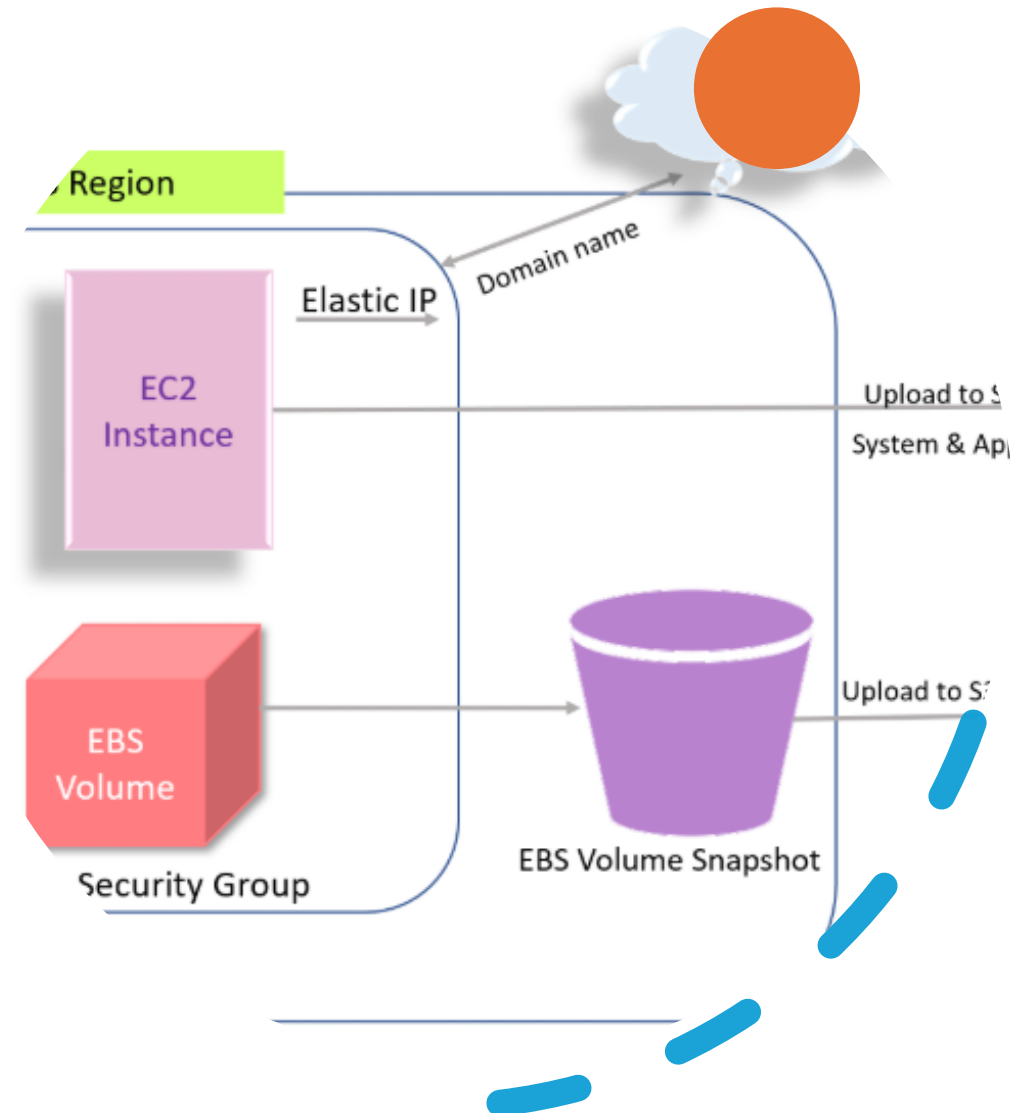
Importance for Cloud Practitioners

- Understanding pricing helps manage AWS resources efficiently.
- Enables cost-effective architecture design.
- Supports proactive monitoring and cost optimization.
- Builds confidence in handling real-world cloud budgeting scenarios.



AWS Architecture and Best Practices

- A set of best practices for designing secure, reliable, efficient, and cost-effective cloud systems
- Helps practitioners build high-quality architectures in the AWS Cloud
- Framework is built on five key pillars



The Five Pillars of the AWS Well-Architected Framework

1. Operational Excellence
 - Focuses on running, monitoring, and improving cloud systems
 - Ensures systems deliver consistent business value
2. Security
 - Protects data, applications, and systems
 - Emphasizes identity management, detection, and data protection
3. Reliability
 - Ensures workloads function correctly and recover from failures
 - Supports system availability and continuity
4. Performance Efficiency
 - Uses computing resources efficiently
 - Adapts to changing demands and new technologies
5. Cost Optimization
 - Avoids unnecessary costs
 - Ensures maximum value through efficient resource use

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Designing for Scalability

- Ensures systems can grow with increasing traffic or workloads
- Improves performance and user experience
- Reduces risk of bottlenecks or service failures



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Tools for Scalability

- Auto Scaling
 - Automatically adjusts compute capacity based on demand
 - Maintains optimal performance while controlling costs
 - Elastic Load Balancing (ELB)
 - Distributes incoming traffic across multiple servers
 - Prevents any single server from becoming overloaded
 - Ensures smooth performance and high availability of applications
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Why It Matters

- Supports reliable, high-performing, and cost-efficient architectures
- Aligns cloud systems with business goals
- Essential knowledge for AWS Cloud Practitioners and Solution Architects



Summary

Cloud Computing

- Delivers computing services (storage, compute, databases) over the Internet
- Offers scalability, flexibility, and pay-as-you-go pricing
- Supports multiple deployment models: Public, Private, Hybrid

AWS Overview

- Leading global cloud provider with 200+ services
- Provides compute (EC2, Lambda), storage (S3, EBS), networking (VPC, Route 53), and databases (RDS, DynamoDB)

AWS Key Principles

- Global infrastructure: Regions & Availability Zones
- Highly scalable, secure, and cost-efficient
- Multiple pricing models: On-Demand, Reserved, Spot
- Cost management tools: AWS Budgets, Cost Explorer

Architecture & Best Practices

- Guided by the AWS Well-Architected Framework (5 pillars: Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization)
- Supports building reliable, scalable, and cost-optimized applications
- Tools like Auto Scaling and Elastic Load Balancing improve performance and availability